Energy barriers and Carleson budgets for off-critical zeros of the Riemann zeta function

Jonathan Washburn

January 2, 2026

Abstract

This paper develops effective near-field barriers against off-critical zeros of $\zeta(s)$ by comparing a quantized windowed phase cost to a scale-tracked Carleson energy budget. It yields an explicit protection height $T_{\mathrm{safe}}(\eta)$ such that any zero $\rho = \beta + i\gamma$ with $\frac{1}{2} < \beta < 0.6$ and $\eta = \beta - \frac{1}{2}$ must satisfy $|\gamma| > T_{\mathrm{safe}}(\eta)$.

1 Introduction

The Riemann zeta function $\zeta(s)$ extends meromorphically to $\mathbb{C}$ with a simple pole at $s = 1$. Its nontrivial zeros encode fine fluctuations of primes, and the Riemann Hypothesis asserts that all nontrivial zeros lie on the critical line $\Re s = \frac{1}{2}$; see [1, 3]. This paper isolates a complementary statement of *effective* type: we derive an explicit height barrier against zeros that lie strictly to the right of the critical line but remain in the near strip

$$\tfrac{1}{2} < \Re s < 0.6.$$

Effective statement (no all-heights claim)

Write a hypothetical off-critical zero as $\rho = \beta + i\gamma$ with $\frac{1}{2} < \beta < 0.6$ and set

$$\eta \;:=\; \beta - \tfrac{1}{2} \in (0, 0.1), \qquad T \;:=\; |\gamma|, \qquad \langle T \rangle \;:=\; \sqrt{1 + T^2}.$$

The main output is a computable *protection height* $T_{\mathrm{safe}}(\eta)$ such that off-critical zeros at depth η cannot occur below height $T_{\mathrm{safe}}(\eta)$. An explicit closed-form definition of $T_{\mathrm{safe}}(\eta)$ (in terms of the budget constants) is given in Definition 8.

Theorem 1 (Effective near-field barrier)**.** *There exists an explicit, computable function $\eta \mapsto T_{\mathrm{safe}}(\eta)$ on $(0, 0.1)$ with the following property: if $\rho = \beta + i\gamma$ is a zero of $\zeta(s)$ with $\frac{1}{2} < \beta < 0.6$ and $\eta = \beta - \frac{1}{2}$, then necessarily*

$$|\gamma| \;>\; T_{\mathrm{safe}}(\eta).$$

Equivalently, for each fixed $\eta \in (0, 0.1)$, there are no zeros of ζ in the region

$$\Big\{ s \in \mathbb{C} : \;\; \Re s > \tfrac{1}{2} + \eta, \;\; \Re s < 0.6, \;\; |\,\Im s\,| \le T_{\mathrm{safe}}(\eta) \Big\}.$$

Remark 2 (Magnitude of the barrier). The bound is extremely large. For example (with the specific window and constants used in this paper), one obtains $T_{\mathrm{safe}}(0.1) \approx 10^{74}$ and $T_{\mathrm{safe}}(0.01) \gtrsim 10^{8800}$. The point of Theorem 1 is therefore not that the resulting height is small, but that it is *fully explicit* and arises from a clean analytic inequality.

Why the result is effective: energy cost versus Carleson budget

The method is an energy comparison with two inputs:

- **Lower bound (quantized phase cost).** An off-critical zero forces a nontrivial amount of boundary phase winding for a normalized zeta-ratio. After localizing by a compactly supported window of width $L \sim 2\eta$, this produces a *minimum* phase-mass cost

$$L_{\mathrm{rec}} \;=\; 4\arctan(2) \;\approx\; 4.428.$$

- **Upper bound (Carleson energy budget).** The same localized phase mass is bounded above by a Carleson-box energy constant for a harmonic extension. This Carleson budget naturally splits into:
  - a *prime-layer* contribution depending only on the local scale L, controlled unconditionally by explicit prime-tail bounds and harmonic-analysis estimates;
  - a *zero-layer* contribution depending on the height T, whose dominant term is proportional to $L\log\langle T\rangle$ and comes from the Poisson balayage of critical-line zeros.

Since the zero-layer term grows with T, the conclusion is inherently height-dependent: the argument rules out an off-critical zero at depth η only up to the point where the available Carleson budget can match the fixed vortex cost. This is the precise sense in which the barrier is *effective* rather than *uniform in height*.

Context within the three-paper split

This paper is the "near-field" component of the split project. Paper I provides an unconditional far-field zero-free region ($\Re s \geq 0.6$) by a certified bounded-real (Schur/Pick) argument. The present paper is logically independent of that certification: it supplies a separate, effective barrier mechanism in the remaining strip $\frac{1}{2} < \Re s < 0.6$.

Roadmap

Section 2 fixes the windowing and phase conventions and defines the Carleson-box energy. Section 3 proves the quantized phase-cost lower bound (the Blaschke trigger), and Section 4 records the CR–Green upper bound by box energy. Section 5 combines these into the barrier inequality and defines $T_{\mathrm{safe}}(\eta)$. Finally, Section 6 discharges the budget decomposition (primes + zeros), completing the effective near-field exclusion.

2 Setup: windowing, phase conventions, and Carleson energy

This section fixes notation for the near-field barrier argument. The main objects are: (i) a compactly supported *window* that localizes near the height of a hypothetical zero, (ii) a boundary *phase* whose distributional derivative records the vortex created by an off-critical zero, and (iii) a *Carleson box energy* that provides an upper budget for that localized phase mass.

Half-plane, boundary parameterization, and off-critical zeros

We work on the right half-plane

$$\Omega \;:=\; \{\, s \in \mathbb{C} :\; \Re s > \tfrac{1}{2} \,\},$$

with boundary line $\partial\Omega = \{\, \frac{1}{2} + it :\; t \in \mathbb{R} \,\}$. Write a hypothetical off-critical zero as $\rho = \beta + i\gamma$ with $\frac{1}{2} < \beta < 0.6$, and set

$$\eta := \beta - \tfrac{1}{2} \in (0, 0.1), \qquad T := |\gamma|.$$

The natural near-field length scale associated to depth η is

$$L := 2\eta, \qquad I_{\gamma,L} := [\gamma - L, \gamma + L].$$

Flat-top windows

Fix once and for all an even C^∞ flat-top cutoff $\psi : \mathbb{R} \to [0,1]$ such that

$$\psi \equiv 1 \;\text{ on } [-1,1], \qquad \operatorname{supp}\psi \subset [-2,2].$$

For $L > 0$ and $t_0 \in \mathbb{R}$ define the rescaled window and its normalized version by

$$\psi_{L,t_0}(t) := \psi\Big(\frac{t - t_0}{L}\Big), \qquad \varphi_{L,t_0}(t) := \frac{1}{L}\,\psi\Big(\frac{t - t_0}{L}\Big), \qquad m_\psi := \int_{\mathbb{R}} \psi(t)\,dt.$$

Then $\int_{\mathbb{R}} \varphi_{L,t_0}(t)\,dt = m_\psi$, the support satisfies $\operatorname{supp}\varphi_{L,t_0} \subset [t_0 - 2L, t_0 + 2L]$, and $\varphi_{L,t_0} \equiv L^{-1}$ on $[t_0 - L, t_0 + L]$. We will test boundary distributions against $\psi_{L,\gamma}$ to localize at the zero's height.

Half-plane Blaschke factors and the phase-cost kernel

For $\rho = \frac{1}{2} + \eta + i\gamma \in \Omega$ define the reflected point across $\partial\Omega$ by

$$\rho^* \;:=\; 1 - \overline{\rho} \;=\; \tfrac{1}{2} - \eta + i\gamma,$$

and define the half-plane Blaschke (pole) factor

$$C_\rho(s) \;:=\; \frac{s - \rho^*}{s - \rho}.$$

On the boundary $s = \frac{1}{2} + it$ one has $|C_\rho(\frac{1}{2} + it)| = 1$ and a direct computation gives the distributional identity

$$\frac{d}{dt} \arg C_\rho\big(\tfrac{1}{2} + it\big) \;=\; \frac{2\eta}{(t-\gamma)^2 + \eta^2} \;\geq\; 0.$$

This is the basic *vortex kernel*: it is nonnegative, concentrated on the scale $|t - \gamma| \lesssim \eta$, and integrates to 2π. Windowing against $\psi_{2\eta,\gamma}$ yields the explicit lower bound

$$\int_{\mathbb{R}} \psi_{2\eta,\gamma}(t)\,\frac{2\eta}{(t-\gamma)^2 + \eta^2}\,dt \;\geq\; 4\arctan(2) \;=:\; L_{\mathrm{rec}}.$$

Phase convention and boundary phase mass

Let $\mathcal{J}$ be a meromorphic function on Ω such that any zero $\rho \in \Omega$ of ζ produces a pole of $\mathcal{J}$ at ρ (for example, a normalized zeta-ratio). On the boundary $\partial\Omega$ we select a phase $w : \mathbb{R} \to \mathbb{R}$ so that the distribution $-w'$ is a *nonnegative* boundary distribution and the contribution of a pole at ρ contains the kernel from the Blaschke factor C_ρ above. Concretely, one may take (where boundary values exist)

$$w(t) \;:=\; -\arg \mathcal{J}(\tfrac{1}{2} + it),$$

i.e. work with $\mathcal{J}^{-1}$ so that pole contributions enter $-w'$ with a positive sign. All phase identities in this paper are used only after testing against a smooth compactly supported window, so $-w'$ is interpreted distributionally.

Carleson boxes and energy

Write points in Ω as $s = \sigma + it$ with $\sigma > \frac{1}{2}$. For an interval $I \subset \mathbb{R}$ and an aperture parameter $\alpha \geq 1$, define the Carleson box above I by

$$Q(\alpha I) \;:=\; I \times (0, \alpha|I|] \;=\; \{\, (\, t, \sigma - \tfrac{1}{2}\,) : \; t \in I, \; 0 < \sigma - \tfrac{1}{2} \leq \alpha|I| \,\}.$$

Let U be a harmonic function on Ω (in the application, U is a log-modulus potential associated to $\mathcal{J}$ on regions where $\mathcal{J}$ is holomorphic and nonvanishing). Define the associated (scale-invariant) energy measure

$$d\lambda \;:=\; |\nabla U(\sigma, t)|^2 \, (\sigma - \tfrac{1}{2}) \, dt \, d\sigma,$$

and the box energy by $\lambda(Q(\alpha I)) = \iint_{Q(\alpha I)} d\lambda$. The Carleson box-energy ratio at base interval I is

$$\mathcal{C}_{\mathrm{box}}(U; I) \;:=\; \frac{1}{|I|} \, \lambda(Q(\alpha I)).$$

At the near-field scale $L = 2\eta$ we will apply this with $I = I_{\gamma,L}$. Later sections establish an explicit upper bound for $\mathcal{C}_{\mathrm{box}}(U; I_{\gamma,L})$ as a function of L and $T = |\gamma|$, and combine it with the windowed lower bound coming from the Blaschke factor to obtain the effective barrier.

3 Phase-cost lower bound (Blaschke trigger)

This section proves the universal lower bound on localized phase mass forced by a single off-critical pole/zero. The estimate is independent of height $T = |\gamma|$ and depends on the depth η only through the localization scale $L = 2\eta$.

Lemma 3 (Blaschke trigger: quantized phase cost)**.** *Let $\rho = \frac{1}{2} + \eta + i\gamma \in \Omega$ with $\eta > 0$ and set $L := 2\eta$. Assume $\mathcal{J}$ is meromorphic on Ω with a pole at ρ and that its boundary phase is chosen as in Section 2 so that the distribution $-w'$ is nonnegative and contains the pole contribution associated to the half-plane Blaschke factor C_ρ. Then one has the windowed lower bound*

$$\int_{\mathbb{R}} \psi_{L,\gamma}(t) \, (-w'(t)) \, dt \;\geq\; L_{\mathrm{rec}}, \qquad L_{\mathrm{rec}} := 4 \arctan(2). \tag{1}$$

Proof. On the boundary line $s = \frac{1}{2} + it$, the pole factor C_ρ satisfies

$$\frac{d}{dt} \arg C_\rho(\tfrac{1}{2} + it) \;=\; \frac{2\eta}{(t-\gamma)^2 + \eta^2} \;\geq\; 0$$

in distributions (Section 2). By hypothesis, $-w'$ is a nonnegative distribution that dominates the pole contribution, hence

$$-w'(t) \;\geq\; \frac{2\eta}{(t-\gamma)^2+\eta^2}$$

in the sense of distributions. Testing against the nonnegative window $\psi_{L,\gamma}$ gives

$$\int_{\mathbb{R}} \psi_{L,\gamma}(t)\,(-w'(t))\,dt \;\geq\; \int_{\mathbb{R}} \psi_{L,\gamma}(t)\,\frac{2\eta}{(t-\gamma)^2+\eta^2}\,dt.$$

Since $L = 2\eta$ and $\psi_{2\eta,\gamma} \equiv 1$ on $[\gamma - 2\eta, \gamma + 2\eta]$, we obtain

$$\int_{\mathbb{R}} \psi_{2\eta,\gamma}(t)\,\frac{2\eta}{(t-\gamma)^2+\eta^2}\,dt \;\geq\; \int_{\gamma-2\eta}^{\gamma+2\eta} \frac{2\eta}{(t-\gamma)^2+\eta^2}\,dt.$$

With the change of variables $u = t - \gamma$, the last integral is explicit:

$$\int_{-2\eta}^{2\eta} \frac{2\eta}{u^2+\eta^2}\,du \;=\; 4\arctan(2) \;=\; L_{\mathrm{rec}}.$$

Combining these inequalities yields (1). □

Remark 4 (Why the trigger is "quantized"). The constant $L_{\mathrm{rec}} = 4\arctan(2)$ is universal: it comes from integrating a fixed fraction of the Poisson kernel associated to the pole at scale $L = 2\eta$. In particular, it does not decay as $T \to \infty$. This height-independence is what makes an energy comparison possible: the lower bound stays fixed while the available Carleson budget varies with scale and height.

4 Carleson energy upper bound (CR–Green / box budget)

This section records the analytic inequality that upper-bounds the same windowed phase mass from Section 3 by a Carleson-box energy. Combined with Lemma 3, it yields the core "cost versus budget" comparison behind the effective barrier.

CR–Green window inequality

Lemma 5 (CR–Green bound for windowed phase mass)**.** *Fix an aperture $\alpha \geq 1$. There exists a constant $C(\psi) = C(\alpha, \psi) < \infty$ with the following property. Let $t_0 \in \mathbb{R}$, $L > 0$, and set the support interval*

$$I^* \;:=\; [t_0 - 2L,\, t_0 + 2L].$$

Assume F is holomorphic and zero-free on the Carleson box $Q(\alpha I^*)$ and write

$$\log F \;=\; U + iV$$

there, with U, V harmonic and V a harmonic conjugate of U. Define the boundary phase $w(t) := -V(\frac{1}{2} + it)$ on I^* (so that $-w'(t) = \partial_t V(\frac{1}{2} + it)$ in distributions). Then

$$\int_{\mathbb{R}} \psi_{L,t_0}(t)\,(-w'(t))\,dt \;\leq\; C(\psi)\,\Big(\iint_{Q(\alpha I^*)} |\nabla U(\sigma,t)|^2\,(\sigma - \tfrac{1}{2})\,dt\,d\sigma\Big)^{1/2}. \tag{2}$$

Proof. Since F is holomorphic and zero-free on $Q(\alpha I^*)$, the branch $\log F$ is holomorphic there. In particular U and V satisfy the Cauchy–Riemann relations on $Q(\alpha I^*)$, and on the boundary line $\Re s = \frac{1}{2}$ we have (in distributions) $\partial_t V = \partial_\sigma U$. Let Ψ_{L,t_0} denote the harmonic (Poisson) extension to Ω of the boundary function ψ_{L,t_0}. With $w(t) = -V(\frac{1}{2} + it)$, we obtain

$$\int_{\mathbb{R}} \psi_{L,t_0}(t)\,(-w'(t))\,dt \;=\; \int_{\mathbb{R}} \psi_{L,t_0}(t)\,\partial_\sigma U(\tfrac{1}{2} + it)\,dt.$$

Green's identity on the box $Q(\alpha I^*)$ (using that both U and Ψ_{L,t_0} are harmonic there, and that ψ_{L,t_0} is supported in I^*) converts this boundary pairing into a Dirichlet pairing of the form

$$\int_{\mathbb{R}} \psi_{L,t_0}(t)\,\partial_\sigma U(\tfrac{1}{2} + it)\,dt \;=\; \iint_{Q(\alpha I^*)} \nabla U \cdot \nabla \Psi_{L,t_0}\,(\sigma - \tfrac{1}{2})\,dt\,d\sigma.$$

Applying Cauchy–Schwarz gives

$$\int_{\mathbb{R}} \psi_{L,t_0}\,(-w') \;\le\; \Big(\iint_{Q(\alpha I^*)} |\nabla U|^2\,(\sigma - \tfrac{1}{2})\Big)^{1/2} \Big(\iint_{Q(\alpha I^*)} |\nabla \Psi_{L,t_0}|^2\,(\sigma - \tfrac{1}{2})\Big)^{1/2}.$$

The second factor is scale-invariant under $(t, \sigma - \frac{1}{2}) \mapsto (t_0 + Lt,\, L(\sigma - \frac{1}{2}))$, hence depends only on α and the fixed window profile ψ. Absorb it into $C(\psi)$ to obtain (2). □

Carleson box formulation

Recall the Carleson ratio from Section 2:

$$\mathcal{C}_{\text{box}}(U; I^*) \;:=\; \frac{1}{|I^*|} \iint_{Q(\alpha I^*)} |\nabla U|^2\,(\sigma - \tfrac{1}{2})\,dt\,d\sigma.$$

Since $|I^*| = 4L$, the bound (2) can be rewritten as

$$\int_{\mathbb{R}} \psi_{L,t_0}(t)\,(-w'(t))\,dt \;\le\; C(\psi)\,\sqrt{|I^*|\,\mathcal{C}_{\text{box}}(U; I^*)} \;=\; C(\psi)\,\sqrt{4L\,\mathcal{C}_{\text{box}}(U; I^*)}. \tag{3}$$

In Sections 5 and 6 we estimate $\mathcal{C}_{\text{box}}(U; I^*)$ for the particular U arising from the zeta-ratio normalization, yielding a computable budget as a function of the near-field scale $L = 2\eta$ and height $T = |t_0|$.

5 Energy barrier inequality and definition of $T_{\text{safe}}(\eta)$

This section packages the near-field argument into a single inequality: any off-critical pole forces a fixed phase-cost (Section 3), while the available phase mass is bounded above by a Carleson energy budget (Section 4). Solving the resulting inequality for height produces the explicit protection height $T_{\text{safe}}(\eta)$.

Barrier inequality: cost versus budget

Proposition 6 (Energy barrier inequality)**.** *Let $\rho = \frac{1}{2} + \eta + i\gamma \in \Omega$ with $\eta \in (0, 0.1)$ and set $L := 2\eta$ and $t_0 := \gamma$. Assume $\mathcal{J}$ has a pole at ρ, and let w be the boundary phase chosen so that $-w'$ is nonnegative and contains the pole contribution (Section 2). Let $I^* = [t_0 - 2L, t_0 + 2L]$ be the support interval of ψ_{L,t_0}.*

If the CR–Green estimate of Lemma 5 applies to a holomorphic/zero-free neutralization of $\mathcal{J}$ on $Q(\alpha I^*)$ with harmonic log-modulus U, then

$$L \cdot \mathcal{C}_{\text{box}}(U; I^*) \;\ge\; \frac{L_{\text{rec}}^2}{4\,C(\psi)^2}. \tag{4}$$

Proof. Apply Lemma 3 with $L = 2\eta$ and $t_0 = \gamma$ to get the lower bound

$$\int_{\mathbb{R}} \psi_{L,t_0}(t)\,(-w'(t))\,dt \;\geq\; L_{\mathrm{rec}}.$$

Apply (3) to the (neutralized) holomorphic/zero-free object on $Q(\alpha I^*)$ to get the upper bound

$$\int_{\mathbb{R}} \psi_{L,t_0}(t)\,(-w'(t))\,dt \;\leq\; C(\psi)\,\sqrt{4L\,\mathcal{C}_{\mathrm{box}}(U;I^*)}.$$

Combining and squaring yields $L_{\mathrm{rec}}^2 \leq 4C(\psi)^2\,L\,\mathcal{C}_{\mathrm{box}}(U;I^*)$, which is (4). □

Budget model and an explicit protection height

To turn (4) into a height exclusion, one needs an explicit upper bound on the box ratio $\mathcal{C}_{\mathrm{box}}(U;I^*)$ in terms of the near-field scale L and height $T := |t_0|$. The guiding structure is a two-term budget:

- a *prime-layer* term depending only on L (height-independent);
- a *zero-layer* term of size $\asymp L\log\langle T\rangle$ coming from the density of critical-line zeros.

Proposition 7 (Carleson budget upper bound)**.** *There exist explicit constants $K_0, K_1, \kappa > 0$ such that for all $L \in (0, 0.2]$ and all heights $T \geq 0$,*

$$\mathcal{C}_{\mathrm{box}}(U;I^*) \;\leq\; \underbrace{K_0 + K_1\log\Bigl(1+\frac{\kappa}{L}\Bigr)}_{\mathcal{C}_{\mathrm{prime}}(L)} \;+\; \underbrace{(1 + L\log\langle T\rangle)}_{\mathcal{C}_{\mathrm{zeros}}(L,T)}, \tag{5}$$

where $I^* = [t_0 - 2L, t_0 + 2L]$ with $|t_0| = T$ and U is the harmonic potential of the neutralized object on $Q(\alpha I^*)$.

The proof of Proposition 7 is given in Section 6.

Definition 8 (Protection height)**.** Let $\eta \in (0, 0.1)$ and set $L := 2\eta$. Define

$$T_{\mathrm{safe}}(\eta) \;:=\; \sqrt{\exp\left(2\,\frac{\frac{L_{\mathrm{rec}}^2}{4\,C(\psi)^2} \;-\; L(\mathcal{C}_{\mathrm{prime}}(L)+1)}{L^2}\right) \;-\; 1}, \qquad \mathcal{C}_{\mathrm{prime}}(L) := K_0 + K_1\log\Bigl(1+\frac{\kappa}{L}\Bigr), \tag{6}$$

with the convention that $T_{\mathrm{safe}}(\eta) = 0$ if the quantity inside the square root is ≤ 0.

Theorem 9 (Effective near-field exclusion from the budget bound)**.** *Assume Proposition 7. If $\rho = \beta + i\gamma$ is a zero of $\zeta(s)$ with $\frac{1}{2} < \beta < 0.6$ and $\eta = \beta - \frac{1}{2}$, then*

$$|\gamma| \;>\; T_{\mathrm{safe}}(\eta),$$

where T_{safe} is defined in (6)*.*

Proof. Let $\rho = \frac{1}{2} + \eta + i\gamma$ be such a zero and set $L := 2\eta$ and $T := |\gamma|$. The barrier inequality (4) gives

$$\frac{L_{\mathrm{rec}}^2}{4\,C(\psi)^2} \;\leq\; L\cdot\mathcal{C}_{\mathrm{box}}(U;I^*).$$

Applying the budget bound (5) yields

$$\frac{L_{\mathrm{rec}}^2}{4\,C(\psi)^2} \;\leq\; L\Bigl(\mathcal{C}_{\mathrm{prime}}(L)+1\Bigr) \;+\; L^2\log\langle T\rangle.$$

Rearranging gives a lower bound on $\log\langle T\rangle$, and the definition (6) is exactly the threshold obtained by solving $\langle T\rangle = \exp(\cdots)$ for T. □

Remark 10 (Why the result is height-limited). The prime-layer term $\mathcal{C}_{\text{prime}}(L)$ is height-independent and grows only logarithmically as $L \downarrow 0$. The obstruction to an all-heights near-field exclusion is the term $L^2 \log\langle T\rangle$: as $T \to \infty$ it eventually dominates at fixed L, so the inequality can no longer contradict the existence of a pole. This is precisely why the present method produces a computable *effective* height bound rather than a uniform zero-free strip.

6 Discharging the budget bound: primes + zeros

This section justifies Proposition 7 by decomposing the Carleson energy into two contributions:

- a **prime-layer term** controlled unconditionally and depending only on the scale L;
- a **zero-layer term** coming from the density of critical-line zeros near height T.

Only standard inputs are used: Mertens-type prime bounds and the Riemann–von Mangoldt zero count; see [2, 3].

A bookkeeping inequality for sums of potentials

In applications, the potential U is built (after local neutralization to ensure harmonicity) as a sum of simpler harmonic potentials. We record a crude but useful estimate.

Lemma 11 (Energy subadditivity up to constants)**.** *Let U_1, U_2 be harmonic on $Q(\alpha I^*)$ and set $U = U_1 + U_2$. Then*

$$\mathcal{C}_{\text{box}}(U; I^*) \;\le\; 2\,\mathcal{C}_{\text{box}}(U_1; I^*) \;+\; 2\,\mathcal{C}_{\text{box}}(U_2; I^*).$$

Proof. Use $|\nabla(U_1 + U_2)|^2 \le 2|\nabla U_1|^2 + 2|\nabla U_2|^2$ pointwise and integrate over $Q(\alpha I^*)$. □

Prime-layer bound (scale-tracked)

The prime-layer term is height-independent and grows only logarithmically as $L \downarrow 0$. We isolate it as a standalone bound.

Proposition 12 (Prime-layer Carleson bound)**.** *There exist absolute constants $K_0, K_1, \kappa > 0$ such that for every $L \in (0, 0.2]$ and every $t_0 \in \mathbb{R}$, the prime-layer contribution satisfies*

$$\mathcal{C}_{\text{prime}}(L) \;\le\; K_0 \;+\; K_1 \log\Big(1 + \frac{\kappa}{L}\Big),$$

where $\mathcal{C}_{\text{prime}}(L)$ denotes the Carleson ratio associated to the prime-controlled part of U on $Q(\alpha I^*)$.

Proof. We indicate the standard bookkeeping behind this bound. At scale L, the prime-controlled part of the potential U is built from a windowed prime-frequency superposition; after expanding its Dirichlet energy on $Q(\alpha I^*)$, one obtains a diagonal term (a weighted ℓ^2 sum over primes) plus an off-diagonal term (a discrete Hilbert-form sum over prime pairs).

The diagonal term produces the logarithmic dependence $\log(1 + \kappa/L)$ by elementary summation together with scale invariance of the box and Mertens-type bounds (see [2, Ch. 1]). The off-diagonal term is controlled by the Montgomery–Vaughan Hilbert inequality (see [2, Ch. 9]), giving at most a constant multiple of the diagonal contribution. Collecting these estimates and absorbing prime tails into K_0 yields the stated bound. □

Zero-layer bound (balayage / zero counting)

The height dependence enters through the density of zeros on the critical line.

Proposition 13 (Zero-layer Carleson bound)**.** *There exists an absolute constant $C > 0$ such that for every $L \in (0, 0.2]$ and every $t_0 \in \mathbb{R}$ (set $T := |t_0|$), the zero-layer contribution satisfies*

$$\mathcal{C}_{\text{zeros}}(L, T) \;\le\; C\Big(1 + L \log\langle T\rangle\Big).$$

Proof. On $Q(\alpha I^*)$ one neutralizes the zeros of ζ (equivalently ξ) in a slightly enlarged box by dividing by the corresponding finite product of half-plane Blaschke factors. The removed factors contribute explicitly and positively to the phase derivative $-w'$; the remaining (neutralized) log-modulus is harmonic on the box. Thus the zero-layer bookkeeping reduces to counting how many critical-line zeros lie at heights comparable to T within an interval of length comparable to L.

By the Riemann–von Mangoldt formula (see [3, Thm. 9.3]), the number of critical-line ordinates in such an interval satisfies

$$N(T + O(L)) - N(T - O(L)) \;\ll\; 1 + L \log\langle T\rangle.$$

Each such zero contributes a uniformly bounded amount to the Carleson ratio at scale L (depending only on the fixed aperture), because the Dirichlet energy of a single half-plane Blaschke factor over $Q(\alpha I^*)$ is $O(|I^*|) \asymp L$. After dividing by $|I^*| \asymp L$ in the definition of $\mathcal{C}_{\text{box}}$, this yields an $O(1)$ contribution per zero, and hence the stated bound. □

Proof of Proposition 7

Proof of Proposition 7. Decompose the (neutralized) potential U on $Q(\alpha I^*)$ into a prime-controlled part plus a zero-layer part. Applying Lemma 11 and absorbing absolute constants into K_0, K_1 and the leading "+1" term gives

$$\mathcal{C}_{\text{box}}(U; I^*) \;\le\; \mathcal{C}_{\text{prime}}(L) \;+\; \mathcal{C}_{\text{zeros}}(L, T).$$

Now apply Proposition 12 and Proposition 13 to obtain (5). □

Conclusion and limitations (effective status)

The argument in this paper is a deterministic energy comparison. An off-critical zero at depth η forces a *quantized* windowed phase cost (the Blaschke trigger, Lemma 3), while the same windowed phase mass is bounded above by a Carleson energy budget via a CR–Green inequality (Lemma 5). This produces the barrier inequality (Proposition 6), and with the explicit two-term budget (Proposition 7) yields an explicit protection height $T_{\text{safe}}(\eta)$ (Definition 8) excluding off-critical zeros below that height (Theorem 9).

What is proved. For each fixed $\eta \in (0, 0.1)$ the method yields a fully explicit height threshold $T_{\text{safe}}(\eta)$ such that no zeros of $\zeta(s)$ can occur in

$$\Big\{ s \in \mathbb{C} : \; \tfrac{1}{2} + \eta < \Re s < 0.6, \; |\,\Im s\,| \le T_{\text{safe}}(\eta) \Big\}.$$

All inputs used to define T_{safe} are classical (prime tail bounds, harmonic analysis on the half-plane, and the Riemann–von Mangoldt formula).

What is *not* proved. This paper does *not* prove a zero-free near-field strip uniformly for all heights, and it does *not* prove the Riemann Hypothesis. The conclusion is explicitly height-limited.

The single obstruction to an all-heights near-field exclusion. The obstruction is transparent in the budget bound (5): the prime-layer term is height-independent, but the zeros term contains the growth

$$\mathcal{C}_{\text{zeros}}(L,T) \asymp 1 + L\log\langle T\rangle.$$

After inserting $L = 2\eta$ into the barrier inequality, this produces the factor $L^2 \log\langle T\rangle$ that eventually dominates as $T \to \infty$ (Remark 10). Any route to an all-heights near-field result within this framework would therefore require a substantially stronger input controlling the zeros contribution uniformly in T.

Statements and Declarations

Competing interests. The author declares no competing interests.

Data and materials availability. This paper contains no new computational artifacts beyond the repository materials used in the unconditional far-field certification (Paper I of this series). All files needed to compile this manuscript are included in the repository.

Reproducibility. All analytic arguments are intended to be reproducible from the stated lemmas and cited sources. When combined with the certified far-field audit of Paper I, the series provides a fully auditable record of the unconditional $\Re s \geq 0.6$ step and the effective near-field barrier mechanism.

References

[1] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, AMS Colloquium Publications, 2004.

[2] H. L. Montgomery and R. C. Vaughan, *Multiplicative Number Theory I: Classical Theory*, Cambridge University Press, 2007.

[3] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., Oxford University Press, 1986.

[4] M. Rosenblum and J. Rovnyak, *Hardy Classes and Operator Theory*, Oxford University Press, 1985.

[5] W. F. Donoghue, *Monotone Matrix Functions and Analytic Continuation*, Springer, 1974.